

Dongjin Kim

AI Researcher · Integrated M.S.–Ph.D. Student
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Summary

Integrated M.S.–Ph.D. student at **KAIST AI** (DAVIAN Lab), advised by Prof. Jaegul Choo. I build **Vision-Language-Action (VLA)** models for robot manipulation, working on how to ground policies in the **spatial structure** of a scene — 3D trajectories, robot-centric pointmaps — building on earlier work localizing sound sources in visual scenes. Alongside this, I work on **LLM safety**: content moderation for specialized domains and defense for LLM agents.

Highlights

- **9 papers at top-tier venues** — CVPR (×2), ICLR (×2), ECCV, EMNLP, IROS, AAAI, ACM MM — with **6 as co-first author**
- Co-first author of **3D HAMSTER** (IROS 2026), a hierarchical VLA for robot manipulation, released with code and model weights
- Work spans robot manipulation (IROS), audio-visual grounding (CVPR ×2, AAAI, ACM MM), and LLM safety (ICLR, EMNLP)

Education

Korea Advanced Institute of Science and Technology (KAIST)

Seongnam, South Korea

Integrated M.S.–Ph.D. in Artificial Intelligence

Mar. 2025 – Present

- DAVIAN Lab, advised by Prof. Jaegul Choo
- Research focus: Vision-Language-Action (VLA), LLM Safety, multi-modal learning

Kyung Hee University

Suwon, South Korea

B.S. in Computer Science & Engineering

Mar. 2019 – Feb. 2025

- Undergraduate Researcher, Visual AI Lab, advised by Prof. Jung Uk Kim
- Research focus: multi-modal learning, video understanding, sound source localization
- Published 4 papers at top venues (CVPR 2024, ACM MM 2023, CVPR 2025, AAAI 2025)

Research Interests

Primary Vision-Language-Action (VLA), LLM Safety

Secondary Multi-modal learning, Video understanding, Sound source localization

Ongoing Research

PointmapVLA: Aligning 3D Perception with Action for Vision-Language-Action Models

In progress

Dongyoon Hwang*, **Dongjin Kim***, Byungkun Lee, Jaegul Choo (*equal contribution)

2026

- Treats 3D as a first-class modality in a VLA rather than an appended encoder, by expressing geometry as robot-frame pointmaps — 3D in image form — that the pretrained visual pathway can read from pretraining onward.

Publications

2026

See like a Robot: Robot-Centric Pointmaps for Vision-Language-Action Models

Under review

Byungkun Lee, Dongyoon Hwang, **Dongjin Kim**, Hojoon Lee, Minho Park, Jaegul Choo

2026

- Robot-centric pointmap representation that supplies scene geometry from the robot's own viewpoint to VLA models.
- Project [arXiv](#)

3D HAMSTER: Bridging Planning and Control in Hierarchical Vision Language Action Models through 3D Trajectory Guidance

IROS 2026

Dongyoon Hwang*, Byungkun Lee*, **Dongjin Kim***, Hyojin Jang, Hoiyeong Jin, Jueun Mun, Minh Park, Hojoon Lee, Hyunseung Kim, Jaegul Choo (*equal contribution)

2026

- Hierarchical VLA that predicts 3D trajectories as an intermediate plan, bridging high-level planning and low-level control for robot manipulation.
- Project [arXiv](#) [Code](#) [Model](#)

MEMBRANE: A Self-Evolving Contrastive Safety Memory for LLM Agent Defense

EMNLP 2026 (Main)

Minseok Choi*, Seungbin Yang*, **Dongjin Kim***, Subin Kim, Jungmin Son, Yunseung Lee, Jaegul Choo, Youngjun Kwak (*equal contribution)

2026

- Self-evolving contrastive safety memory that defends LLM agents against jailbreaks without retraining.
- [arXiv](#)

ExpGuard: LLM Content Moderation in Specialized Domains

ICLR 2026

Minseok Choi*, **Dongjin Kim***, Seungbin Yang, Subin Kim, Youngjun Kwak, Juyoung Oh, Jaegul Choo, Jungmin Son (*equal contribution)

2026

- Content moderation for LLMs in specialized domains, where general-purpose safety classifiers transfer poorly.
- [arXiv](#) [OpenReview](#) [Code](#)

LiveWeb-IE: A Benchmark For Online Web Information Extraction

ICLR 2026

Seungbin Yang, Jihwan Kim, Jaemin Choi, **Dongjin Kim**, Soyoung Yang, ChaeHun Park, Jaegul Choo

2026

- Benchmark for information extraction from live web pages, evaluating models on continually changing online content.
- [arXiv](#) [OpenReview](#)

InsertAnywhere: Geometrically Grounded and Optics-Aware Video Object Insertion

ECCV 2026

Hoiyeong Jin, Hyojin Jang, Junha Hyung, Jeongho Kim, Kinam Kim, **Dongjin Kim**, Huijin Choi, Hyeonji Kim, Jaegul Choo

2026

- Video object insertion that propagates a user-specified placement across frames using reconstructed 4D scene geometry, with a diffusion model matching scene optics.
- Project [arXiv](#) [Code](#)

2025

Object-aware Sound Source Localization via Audio-Visual Scene Understanding

CVPR 2025

Sung Jin Um*, **Dongjin Kim***, Sangmin Lee, Jung Uk Kim (*equal contribution)

2025

- Object-aware sound source localization that grounds audio in visual scenes via audio-visual scene understanding.
- [arXiv](#) [Code](#)

Watch Video, Catch Keyword: Context-aware Keyword Attention for Moment Retrieval and Highlight Detection

AAAI 2025

Sung Jin Um, **Dongjin Kim**, Sangmin Lee, Jung Uk Kim

2025

- Context-aware keyword attention for joint moment retrieval and highlight detection in video.
- [arXiv](#) [Code](#)

TV-LiVE: Training-Free, Text-Guided Video Editing via Layer Informed Vitality Exploitation

arXiv 2025

Min-Jung Kim, **Dongjin Kim**, Seokju Yun, Jaegul Choo

2025

- Training-free, text-guided video editing that exploits layer-wise vitality in a pretrained video diffusion model.
- Project [arXiv](#)

2024

Learning to Visually Localize Sound Sources from Mixtures without Prior Source Knowledge

CVPR 2024

Dongjin Kim*, Sung Jin Um*, Sangmin Lee, Jung Uk Kim (*equal contribution)

2024

- Visually localizes multiple sound sources from audio mixtures without prior knowledge of the constituent sources.
- arXiv Code

2023

Audio-Visual Spatial Integration and Recursive Attention for Robust Sound Source Localization

ACM MM 2023

Sung Jin Um*, Dongjin Kim*, Jung Uk Kim (*equal contribution)

2023

- Audio-visual spatial integration with recursive attention for sound source localization robust to noisy scenes.
- arXiv Code

Experience

KAIST AI, DAVIAN Lab

Seongnam, South Korea

Graduate Researcher

Mar. 2025 – Present

- Research on VLA, LLM Safety, and multi-modal learning
- Advised by Prof. Jaegul Choo

LETSUR (AI Startup)

Seoul, South Korea

AI Engineer

Mar. 2024 – Aug. 2024

- Development of AI-based services utilizing LLMs and RAG

ETRI (Electronics and Telecommunications Research Institute)

Daejeon, South Korea

AI Research Intern

Jan. 2024 – Feb. 2024

- Research on Video Moment Retrieval and Highlight Detection

Kyung Hee University, Visual AI Lab

Suwon, South Korea

Undergraduate Researcher

Nov. 2022 – Feb. 2025

- Supervised by Prof. Jung Uk Kim